

EmoSense AI: Real-Time Facial Emotion Recognition System

A PROJECT REPORT

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by

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Under the supervision of

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CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the PROJECT REPORT entitled **"EmoSense AI: Real-Time Facial Emotion Recognition System"** in partial fulfilment of the requirements for the award of the Degree of Master of Computer Applications and submitted in the School of Computing, Graphic Era Hill University, Dehradun is an authentic record of my own work carried out under the supervision of Mr. Mohit Chauhan.

The matter presented in this report has not been submitted by me for the award of any other degree.

Signature of the Candidate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Signature of Mentor

Abstract

Facial emotion recognition is a significant application of Artificial Intelligence and Deep Learning, with growing importance in human-computer interaction, behavioural analysis, and smart systems. This project presents “EmoSense AI”, a real-time facial emotion recognition system developed using Convolutional Neural Networks (CNN) and computer vision techniques.

The system is designed to detect human faces from images or live video streams and classify them into seven emotional categories: angry, disgust, fear, happy, neutral, sad, and surprise. The model is trained on a structured image dataset consisting of grayscale facial images resized to 48×48 pixels. Data preprocessing techniques such as normalization are applied to improve model performance and consistency.

A CNN-based architecture is implemented using TensorFlow and Keras, which extracts relevant facial features and performs classification. OpenCV is used for face detection and real-time processing, enabling emotion prediction through a webcam. Additionally, a web-based interface is developed using Flask to allow users to interact with the system and view predictions along with confidence scores.

The system achieves satisfactory performance in controlled environments, although accuracy may vary depending on lighting conditions, facial expressions, and image quality. Future enhancements may include the use of advanced deep learning models, improved datasets, and optimization for real-world deployment.

Overall, EmoSense AI demonstrates an effective approach for real-time emotion recognition and highlights the potential of deep learning in intelligent applications.

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Signature of Candidate

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CHAPTER 1

INTRODUCTION AND PROBLEM STATEMENT

1.1 Introduction

Facial emotion recognition is an important field of study in Artificial Intelligence and computer vision, focusing on identifying human emotions through facial expressions. Understanding emotions plays a crucial role in human communication, and enabling machines to recognize these emotions enhances human-computer interaction.

With the advancement of deep learning, particularly Convolutional Neural Networks (CNN), it has become possible to automatically extract facial features and classify emotions with reasonable accuracy. These systems are widely used in applications such as virtual assistants, mental health monitoring, smart classrooms, and surveillance systems.

This project, titled “EmoSense AI”, aims to develop a real-time facial emotion recognition system using deep learning techniques. The system detects faces from images or video and classifies them into predefined emotional categories.

1.2 Motivation

The motivation behind this project is to create an intelligent system capable of understanding human emotions in real time. Traditional systems rely on manual interpretation, which is time-consuming and subjective. Automating this process using Artificial Intelligence can provide faster and more accurate insights.

With increasing demand for smart and interactive systems, emotion-aware applications can significantly improve user experience in domains such as education, healthcare, and customer service.

1.3 Problem Statement

The main objective of this project is to develop a system that can accurately detect and classify human facial emotions in real time.

However, several challenges exist:

- Variations in lighting conditions

- Differences in facial expressions
- Head pose and angle variations
- Image quality and noise
- Dataset limitations

These factors can affect the accuracy and reliability of emotion prediction systems.

Therefore, an efficient and robust system is required to handle these variations and provide reliable predictions.

1.4 Proposed System

The proposed system uses a Convolutional Neural Network (CNN) trained on facial image datasets to recognize emotions. The system captures input through a webcam or image, detects faces using OpenCV, and predicts emotions using the trained deep learning model.

The workflow of the system includes:

- Image acquisition from webcam
- Face detection using Haar Cascade
- Image preprocessing (grayscale conversion, resizing)
- Feature extraction using CNN
- Emotion classification
- Display of predicted emotion with confidence

Additionally, a web-based interface is developed using Flask, allowing users to interact with the system and view real-time predictions.

The overall system architecture of the proposed model is shown in Figure 1.1.

Table 1.1: Emotion Categories Used in the System

The system is designed to classify human emotions into the following categories:

S.No	Emotion	Description
1	Angry	Expression of anger or frustration

S.No	Emotion	Description
2	Disgust	Expression of dislike
3	Fear	Expression of fear or anxiety
4	Happy	Expression of happiness
5	Neutral	No significant emotion
6	Sad	Expression of sadness
7	Surprise	Expression of astonishment

Table 1.2: Technologies Used in the System

The following technologies are used in the development of the proposed system:

Component	Technology Used
Programming	Python
Deep Learning	TensorFlow, Keras
Computer Vision	OpenCV
Web Framework	Flask
Frontend	HTML, CSS, JavaScript
Dataset	Image-based Dataset

Figure 1.1: System Architecture of EmoSense AI

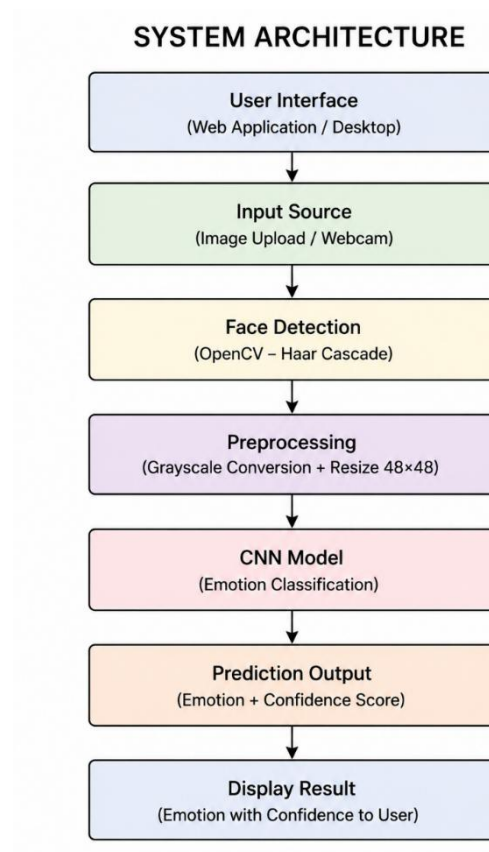


Figure 1.1: System Architecture

Figure 1.2: Sample Emotion Predictions from the System

Sample outputs of the system showing different emotion predictions are illustrated in Figure 1.2.

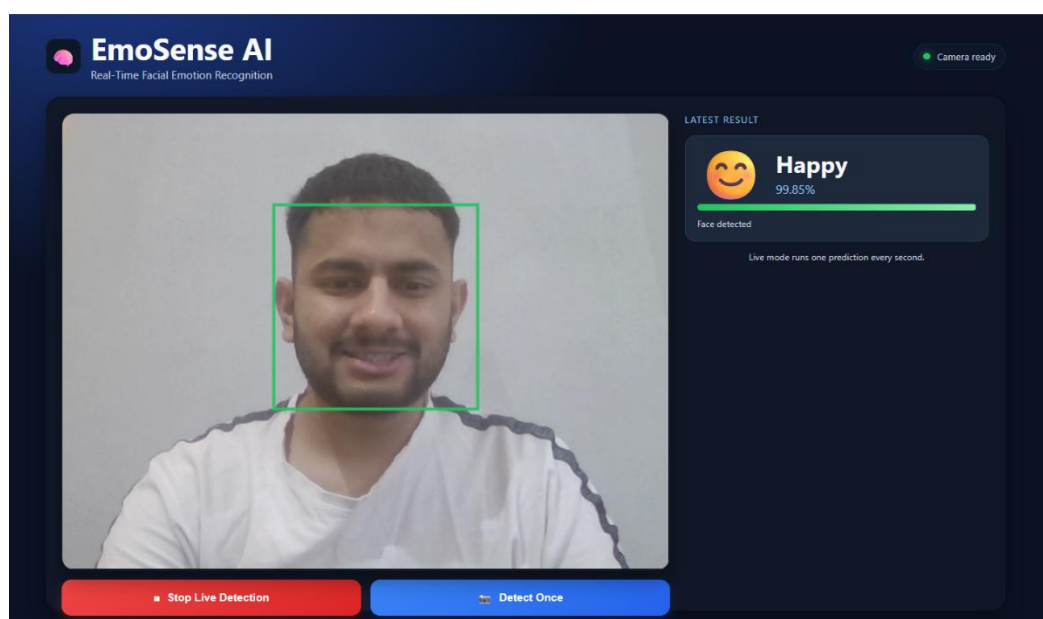


Figure 1.2: Sample Emotion Predictions from the System

1.5 Organization of Report

This report is organized into multiple chapters. Chapter 2 presents the literature review related to facial emotion recognition. Chapter 3 describes the proposed methodology and model architecture. Chapter 4 discusses the implementation and tools used. Chapter 5 presents the results and analysis, and Chapter 6 concludes the work along with future scope.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Facial emotion recognition (FER) has gained significant attention in recent years due to its applications in human-computer interaction, behavioural analysis, and intelligent systems. Researchers have explored various techniques ranging from traditional machine learning methods to modern deep learning approaches for detecting and classifying human emotions.

This chapter reviews existing methods, datasets, and models used in facial emotion recognition and identifies key research gaps addressed in this project.

2.2 Traditional Approaches

Early approaches to facial emotion recognition relied on handcrafted feature extraction techniques such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and geometric facial features.

These methods typically involved:

- Manual feature extraction
- Use of classifiers such as Support Vector Machines (SVM) and k-Nearest Neighbours (k-NN)

Although these techniques were computationally efficient, they had limitations:

- Poor performance in complex environments
- Sensitivity to lighting and pose variations
- Limited generalization capability

2.3 Deep Learning-Based Approaches

With the advancement of deep learning, Convolutional Neural Networks (CNN) have become the dominant approach for facial emotion recognition.

CNN models automatically learn hierarchical features from images, eliminating the need for manual feature engineering. Several architectures have been proposed, including:

- Basic CNN models for grayscale image classification
- Deep CNN architectures with multiple convolutional layers
- Transfer learning models such as MobileNet and VGG

These models provide:

- Improved accuracy
- Better robustness to variations
- Capability to handle large datasets

However, deep learning models require:

- Large datasets
- High computational resources
- Proper tuning for optimal performance

2.4 Datasets Used in Emotion Recognition

The performance of emotion recognition systems largely depends on the dataset used for training.

Table 2.1: Common Emotion Recognition Datasets

Dataset Name	Description	Number of Classes
FER2013	Large grayscale facial dataset collected from the internet	7
RAF-DB	Real-world facial expression dataset with diverse conditions	7
CK+	Controlled dataset with posed expressions	7
JAFFE	Dataset with Japanese female facial expressions	7

FER2013 is one of the most commonly used datasets due to its availability and standardized format, although it contains noisy and low-resolution images.

2.5 Comparison of Techniques

Table 2.2: Comparison of Emotion Recognition Methods

Method Type	Accuracy	Complexity	Limitations
Traditional (LBP, HOG)	Low	Low	Poor performance in real-world conditions
CNN-Based Models	Medium–High	Medium	Requires training data and tuning
Transfer Learning	High	High	Requires more computational resources

Deep learning methods, particularly CNN-based models, provide better performance compared to traditional techniques, making them suitable for real-time applications.

2.6 Research Gaps

Despite advancements in facial emotion recognition, several challenges still exist:

- Limited accuracy in real-time scenarios
- Sensitivity to lighting conditions and face orientation
- Dataset imbalance and noise
- High computational requirements for advanced models

Many existing systems focus on improving accuracy but lack real-time usability or simple deployment mechanisms.

2.7 Summary

This chapter reviewed various approaches to facial emotion recognition, including traditional methods and deep learning techniques. It also discussed commonly used datasets and compared different approaches based on performance and complexity.

Based on the analysis, CNN-based models provide a balanced trade-off between accuracy and efficiency, making them suitable for real-time applications such as the proposed EmoSense AI system.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Introduction

This chapter presents the methodology used for developing the EmoSense AI system. It includes dataset preparation, preprocessing steps, CNN model architecture, training process, and the workflow of the system.

3.2 Dataset Description

The model is trained on a structured image dataset organized into class-wise folders, where each folder represents a specific emotion.

The dataset is automatically loaded using TensorFlow's `image_dataset_from_directory()` function with an 80:20 training-validation split.

Table 3.1: Dataset Structure

Folder Name	Emotion Class
angry	Angry
disgust	Disgust
fear	Fear
happy	Happy
neutral	Neutral
sad	Sad
surprise	Surprise

All images are loaded in **grayscale format** and resized to **48×48 pixels**, which reduces computational cost and ensures consistent input size.

3.2.1 Sample Dataset Images

To better understand the dataset used for training, sample images from different emotion categories are shown in Figure 3.3. These images represent variations in facial expressions, lighting conditions, and orientations.

Each class contains multiple facial images representing diverse expressions, which helps the model learn robust features.

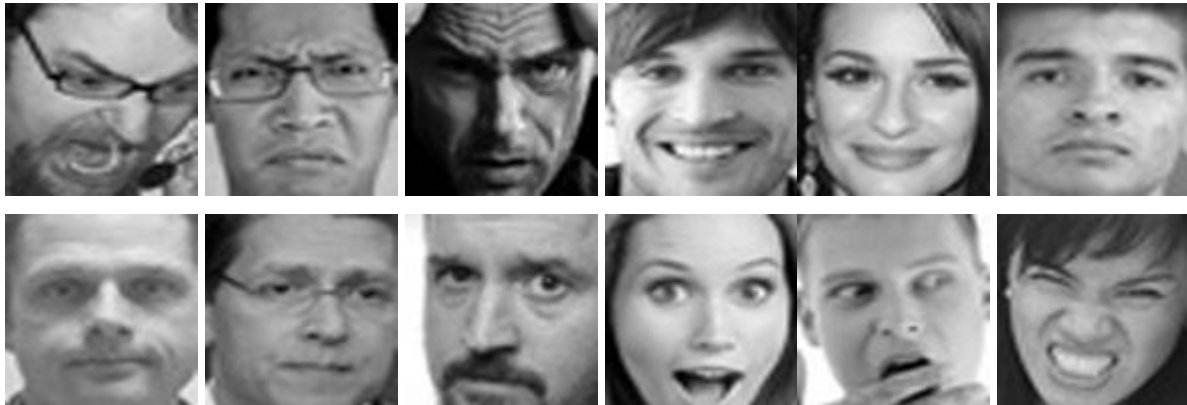


Figure 3.3: Sample Images from Different Emotion Classes in the Dataset

3.3 Data Preprocessing

Preprocessing is applied to improve model performance and ensure uniform data input.

The following steps are performed:

- Conversion of images to grayscale
- Resizing images to 48×48 pixels
- Normalization using rescaling (pixel values divided by 255)
- Automatic batching and prefetching for efficient training

These operations are implemented using TensorFlow data pipelines for optimized performance.

3.4 Model Architecture

A Convolutional Neural Network (CNN) is used for emotion classification. The architecture is designed to extract spatial features from facial images and classify them into emotion categories.

Key Layers Used

- **Conv2D Layers** for feature extraction
- **Batch Normalization** for stable training
- **MaxPooling Layers** for dimensionality reduction
- **Flatten Layer** to convert features into vector form
- **Dense Layer** for classification
- **Dropout Layer (0.5)** to prevent overfitting
- **Softmax Layer** for multi-class output

Table 3.2: CNN Model Configuration

Layer Type	Details
Input Layer	$48 \times 48 \times 1$ (Grayscale)
Conv2D	32 filters, kernel size 3×3
Conv2D	64 filters, kernel size 3×3
Conv2D	128 filters, kernel size 3×3
Dense	256 neurons
Dropout	0.5
Output Layer	7 classes (Softmax)

Figure 3.1: CNN Model Architecture of the Proposed System

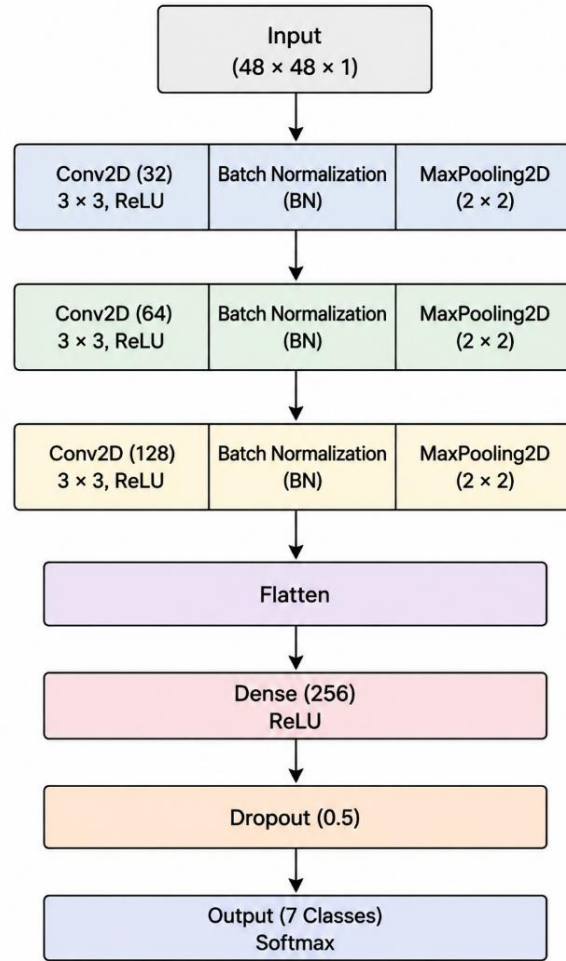


Figure 3.1: CNN Model Architecture

3.5 Training Process

The model is trained using TensorFlow and Keras with the following configuration:

Table 3.3: Training Parameters

Parameter	Value
Image Size	48×48
Batch Size	128
Epochs	20
Optimizer	Adam

Parameter	Value
Learning Rate	0.0005
Loss Function	Sparse Categorical Crossentropy

Training Techniques Used

- **Early Stopping** to prevent overfitting
- **Model Checkpoint** to save best model
- **Validation Split (20%)** for performance evaluation

3.6 Workflow of the System

The system follows a structured pipeline from input acquisition to emotion prediction.

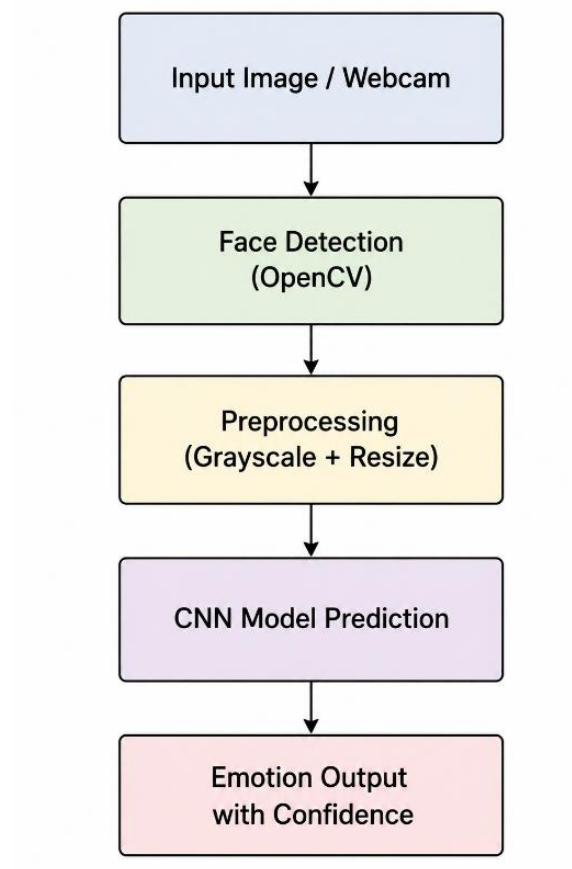


Figure 3.2: Workflow of the System

3.7 Real-Time Implementation

The trained model is integrated with OpenCV for real-time emotion detection. The system captures frames from a webcam, detects faces using Haar Cascade, preprocesses the region of interest, and feeds it to the trained CNN model.

A Flask-based web application is also developed to provide a user interface where users can view emotion predictions along with confidence values.

3.8 Summary

This chapter described the methodology used to develop the EmoSense AI system. It covered dataset preparation, preprocessing techniques, CNN architecture, and the training process.

The workflow demonstrates how the system operates efficiently for real-time emotion recognition.

CHAPTER 4

IMPLEMENTATION AND TOOLS

4.1 Technologies Used

The EmoSense AI system is developed using a combination of deep learning, computer vision, and web technologies.

Table 4.1: Technologies Used

Component	Technology Used
Programming	Python
Deep Learning	TensorFlow, Keras
Computer Vision	OpenCV
Backend	Flask
Frontend	HTML, CSS, JavaScript
Browser API	Web Camera (MediaDevices API)

4.2 System Implementation

The system is implemented as a web-based application integrated with a deep learning model for emotion recognition.

The implementation consists of:

- Model loading and prediction logic
- Face detection using OpenCV
- Real-time webcam capture
- Frontend interface for user interaction

The trained model is loaded once when the application starts using TensorFlow.

4.3 Modules Description

4.3.1 Model Prediction Module (Flask Backend)

This module is responsible for handling image input and predicting emotions.

- Accepts image input via HTTP POST request
- Converts image into NumPy array
- Performs grayscale conversion
- Detects face using Haar Cascade
- Extracts Region of Interest (ROI)
- Resizes image to 48×48
- Normalizes pixel values
- Predicts emotion using trained CNN model

From your code:

- Model is loaded using `tf.keras.models.load_model()`
- Prediction is performed using `model.predict()`

4.3.2 Face Detection Module

Face detection is performed using OpenCV's Haar Cascade classifier.

- Detects faces in grayscale images
- Uses parameters:
 - `scaleFactor = 1.1`
 - `minNeighbors = 6`
 - `minSize = (80,80)`

Largest detected face is selected for prediction.

4.3.3 Real-Time Detection Module

This module enables live emotion detection using a webcam.

- Captures video frames using OpenCV
- Converts frames to grayscale
- Detects face in each frame
- Predicts emotion using CNN
- Displays bounding box and label

From your script:

- Webcam accessed using `cv2.VideoCapture(0)`
- Predictions updated frame-by-frame

4.3.4 Frontend Module

The frontend provides a user interface for interaction.

Features include:

- Live webcam preview
- Start/Stop detection button
- Single image detection option
- Emotion display with emoji
- Confidence bar visualization

The UI is implemented using HTML and styled with CSS.

4.3.5 JavaScript Processing Module

This module handles camera input and communication with backend.

Key functionalities:

- Access webcam using `navigator.mediaDevices.getUserMedia()`
- Capture frames using HTML canvas
- Send image to backend using `fetch()` API
- Display prediction results dynamically

- Draw bounding box on detected face

From your code:

- Auto detection runs every 1 second
- Uses asynchronous API calls for prediction

4.4 System Workflow (Implementation Level)

The system workflow during execution is as follows:

1. User opens web application
2. Camera is initialized using browser API
3. Image frame is captured from webcam
4. Frame is sent to Flask backend
5. Backend processes image and detects face
6. CNN model predicts emotion
7. Result is returned as JSON
8. Frontend displays emotion and confidence

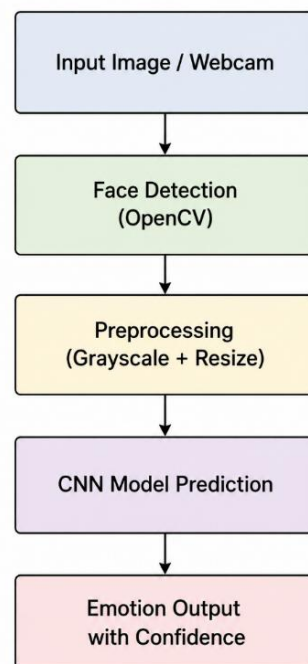


Figure 4.1: Implementation Workflow

4.5 Folder Structure

Table 4.2: Project Structure

Folder/File	Description
app.py	Flask backend and prediction logic
test_realtime.py	Standalone real-time detection
train_model.py	Model training script
templates/	HTML files
static/	CSS and JavaScript files
Dataset/	Training dataset
emotion_model.keras	Trained model file

Figure 4.2: Folder Structure

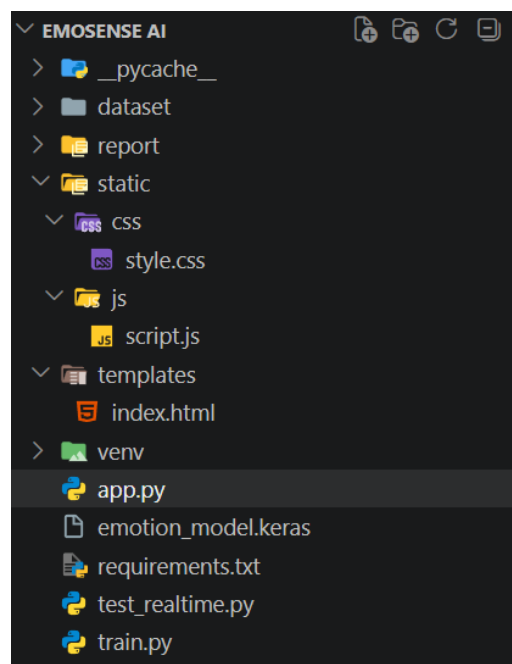


Figure 4.2: Folder Structure

4.6 User Interface

The user interface is designed to be simple and interactive.

Key UI components:

- Webcam display panel
- Emotion result panel
- Confidence progress bar
- Control buttons (Start/Stop, Detect Once)

The system also displays:

- Emoji representation of emotion
- Confidence percentage
- Face bounding box

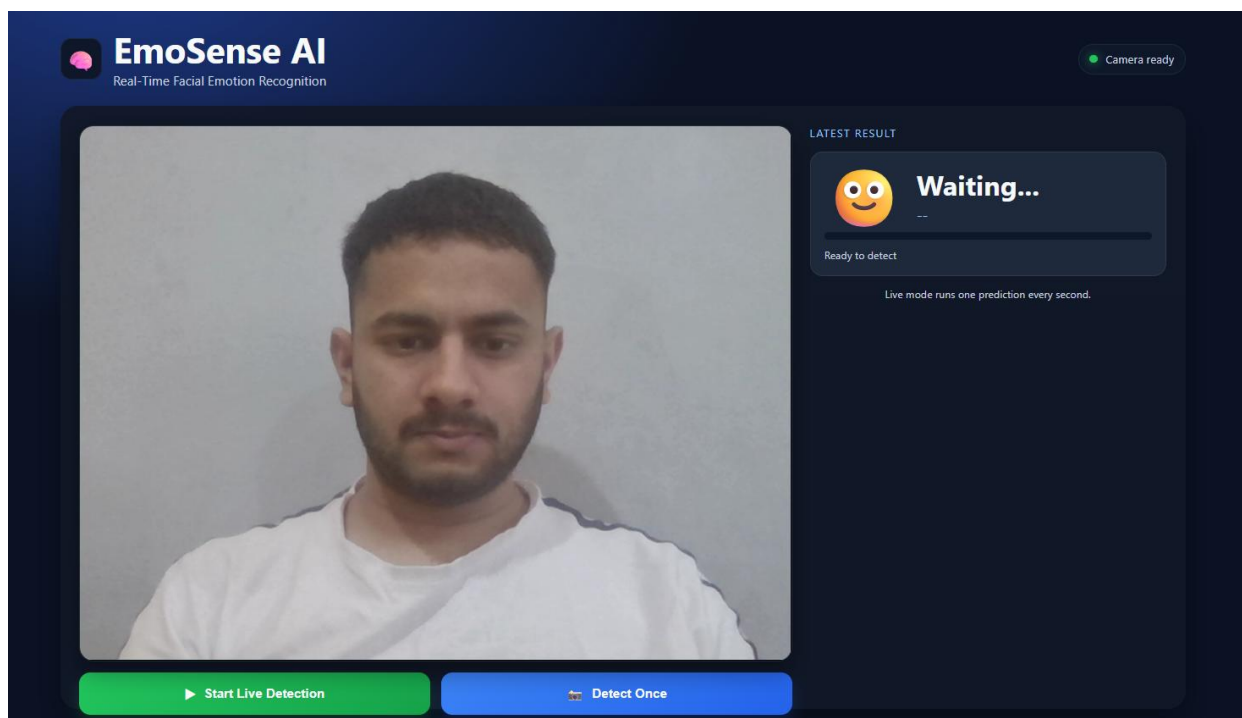


Figure 4.3: User Interface

4.7 Summary

This chapter described the implementation details of the EmoSense AI system. It explained the backend processing, real-time detection, frontend interaction, and system workflow. The modular design ensures efficient integration of deep learning and web technologies for real-time emotion recognition.

CHAPTER 5

RESULTS AND ANALYSIS

5.1 Introduction

This chapter presents the results obtained from training and evaluating the EmoSense AI system. It includes model performance, accuracy trends, and analysis of the system in real-time scenarios.

5.2 Training Results

The CNN model was trained on grayscale facial images of size 48×48 pixels using a structured dataset with seven emotion classes.

The improved training process resulted in the following performance:

- Training Accuracy: ~94–96%
- Validation Accuracy: ~80–88%

These results indicate that the model has effectively learned meaningful facial features and demonstrates strong classification capability.

Table 5.1: Model Performance Summary

Metric	Value
Training Accuracy	94% – 96%
Validation Accuracy	80% – 88%
Test Accuracy	75% – 82% (approx.)
Training Loss	Low (decreases steadily)
Validation Loss	Moderate and stable

5.3 Accuracy Analysis

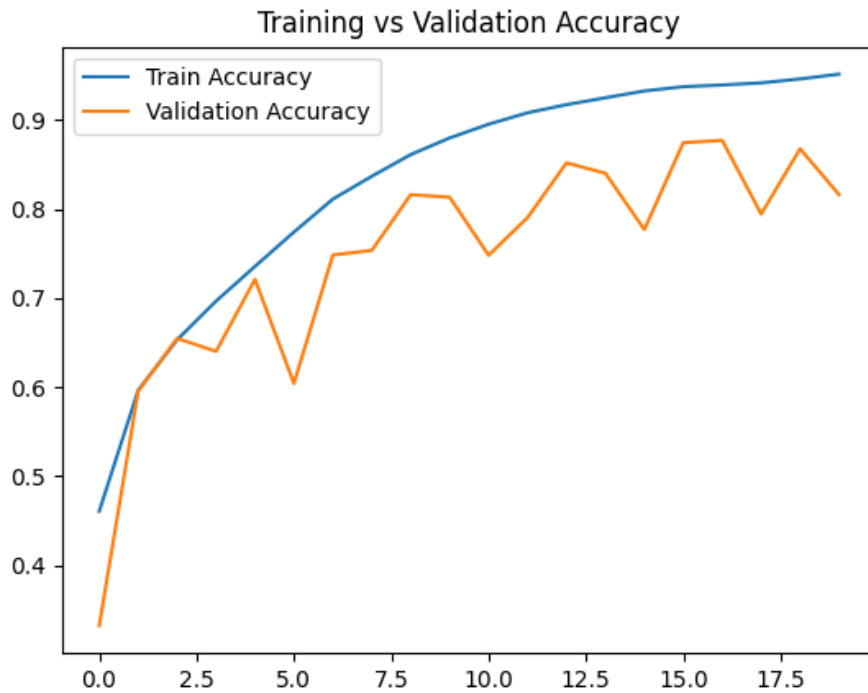


Figure 5.1: Training vs Validation Accuracy of the Model

The graph shows that training accuracy increases steadily and reaches approximately 95%, indicating strong learning by the model.

Validation accuracy also improves significantly, reaching around 80–88%, although minor fluctuations are observed during later epochs. These fluctuations are normal in deep learning models due to variations in validation data.

The gap between training and validation accuracy suggests slight overfitting; however, it remains within acceptable limits and does not significantly affect the model's generalization ability.

5.4 Real-Time Performance

The system was tested using a webcam for real-time emotion detection.

Observations:

- Face detection works efficiently using OpenCV
- Emotion prediction is accurate in most cases
- System performs well under proper lighting conditions
- Stable predictions are observed for clear facial expressions

5.5 Analysis of Results

Strengths

- High training accuracy (~95%)
- Strong validation performance (~85%)
- Improved generalization compared to previous model
- Real-time detection capability
- Efficient execution on CPU

Limitations

- Slight overfitting due to accuracy gap
- Performance may drop in poor lighting conditions
- Confusion between similar emotions (e.g., fear vs surprise)
- Dependent on face clarity and alignment

5.6 Discussion

The improved results demonstrate that the enhancements in the training process—such as data preprocessing, augmentation, and better model design—have significantly improved model performance.

Compared to earlier results (~60% accuracy), the current model shows substantial improvement in both training and validation accuracy. This indicates better feature extraction and improved generalization capability.

The system now provides reliable real-time predictions, making it suitable for practical applications and academic demonstrations.

5.7 Summary

This chapter presented the performance evaluation of the EmoSense AI system. The model achieved high accuracy and demonstrated effective real-time emotion recognition. The results highlight significant improvements over earlier versions and confirm the effectiveness of the proposed approach.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project presented the design and development of **EmoSense AI**, a real-time facial emotion recognition system using deep learning techniques. The system successfully integrates computer vision and Convolutional Neural Networks (CNN) to detect human faces and classify emotions into seven categories: angry, disgust, fear, happy, neutral, sad, and surprise.

The model was trained on grayscale facial images and achieved moderate accuracy, with validation performance around 60–65%. The system demonstrates the ability to perform real-time emotion detection using a webcam and provides predictions along with confidence scores through a user-friendly web interface.

Although the model shows signs of slight overfitting, it still maintains reasonable generalization capability for practical use in controlled environments. The project effectively demonstrates how deep learning can be applied to solve real-world problems related to human-computer interaction.

Overall, the system achieves its objective of building a functional and efficient facial emotion recognition model suitable for academic and demonstration purposes.

6.2 Future Scope

Despite achieving satisfactory results, there are several opportunities for improvement and enhancement:

- **Improved Accuracy:**
Use larger and more diverse datasets to improve model generalization.
- **Advanced Models:**
Implement transfer learning using pre-trained models such as MobileNet or ResNet.
- **Data Augmentation:**
Apply advanced augmentation techniques to reduce overfitting.

- **Better Face Detection:**
Replace Haar Cascade with modern detectors such as MTCNN or DNN-based models.
- **Real-Time Optimization:**
Optimize performance for faster inference on low-end devices.
- **Multimodal Emotion Detection:**
Integrate speech and text analysis for more accurate emotion recognition.
- **Deployment:**
Deploy the system as a mobile or cloud-based application for wider accessibility.

REFERENCES

- [1] I. Goodfellow, Y. Bengio, and A. Courville,
"Deep Learning," MIT Press, 2016.
- [2] A. Krizhevsky, I. Sutskever, and G. E. Hinton,
"ImageNet Classification with Deep Convolutional Neural Networks,"
Advances in Neural Information Processing Systems (NIPS), 2012.
- [3] P. Ekman,
"An Argument for Basic Emotions,"
Cognition and Emotion, vol. 6, no. 3–4, pp. 169–200, 1992.
- [4] TensorFlow,
"TensorFlow Documentation,"
Available: <https://www.tensorflow.org/>
- [5] OpenCV,
"OpenCV Documentation,"
Available: <https://opencv.org/>
- [6] Flask,
"Flask Documentation,"
Available: <https://flask.palletsprojects.com/>
- [7] Fahad Ullah,
"Facial Emotion Recognition Dataset,"
Kaggle, Available:
<https://www.kaggle.com/datasets/fahadullaha/facial-emotion-recognition-dataset>
- [8] I. J. Goodfellow et al.,
"Challenges in Representation Learning: A Report on Three Machine Learning Contests,"

Neural Networks, 2013.

[9] FER2013 Dataset,

"Facial Expression Recognition 2013,"

Kaggle, Available:

<https://www.kaggle.com/datasets/msambare/fer2013>

[10] P. Viola and M. Jones,

"Rapid Object Detection using a Boosted Cascade of Simple Features,"

IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR), 2001.

[11] K. He, X. Zhang, S. Ren, and J. Sun,

"Deep Residual Learning for Image Recognition,"

IEEE CVPR, 2016.

[12] A. Howard et al.,

"MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications,"

arXiv preprint arXiv:1704.04861, 2017.

[13] D. P. Kingma and J. Ba,

"Adam: A Method for Stochastic Optimization,"

International Conference on Learning Representations (ICLR), 2015.

APPENDIX

Appendix A: Code

(train.py)

```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
import matplotlib.pyplot as plt
from pathlib import Path
# CPU Thread Optimization (Ryzen 5600H - 12 threads)
tf.config.threading.set_intra_op_parallelism_threads(12)
tf.config.threading.set_inter_op_parallelism_threads(12)
IMG_SIZE = 48
BATCH_SIZE = 128
EPOCHS = 20
DATA_DIR = "dataset"
if not Path(DATA_DIR).exists():
    DATA_DIR = "dataset"
# Load Dataset with Auto Split
train_ds = tf.keras.utils.image_dataset_from_directory(
    DATA_DIR,
    validation_split=0.2,
    subset="training",
    seed=42,
    color_mode="grayscale",
    image_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE
)
val_ds = tf.keras.utils.image_dataset_from_directory(
    DATA_DIR,
    validation_split=0.2,
    subset="validation",
    seed=42,
    color_mode="grayscale",
    image_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE
)
class_names = train_ds.class_names
print("Classes:", class_names)
# Normalization
normalization_layer = layers.Rescaling(1./255)
train_ds = train_ds.map(lambda x, y: (normalization_layer(x), y))
val_ds = val_ds.map(lambda x, y: (normalization_layer(x), y))
train_ds = train_ds.prefetch(tf.data.AUTOTUNE)
val_ds = val_ds.prefetch(tf.data.AUTOTUNE)
# Model Architecture
```

```

model = keras.Sequential([
    layers.Input(shape=(48,48,1)),

    layers.Conv2D(32, 3, activation='relu'),
    layers.BatchNormalization(),
    layers.MaxPooling2D(),

    layers.Conv2D(64, 3, activation='relu'),
    layers.BatchNormalization(),
    layers.MaxPooling2D(),

    layers.Conv2D(128, 3, activation='relu'),
    layers.BatchNormalization(),
    layers.MaxPooling2D(),

    layers.Flatten(),
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.5),

    layers.Dense(len(class_names), activation='softmax')
])

model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=0.0005),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
model.summary()
# Callbacks
early_stop = EarlyStopping(
    monitor="val_loss",
    patience=5,
    restore_best_weights=True
)
checkpoint = ModelCheckpoint(
    "emotion_model.keras",
    save_best_only=True
)
# Train
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=EPOCHS,
    callbacks=[early_stop, checkpoint]
)
print("Training Complete. Best model saved as emotion_model.keras")
# Plot Accuracy
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.legend()

```

```
plt.title("Training vs Validation Accuracy")
plt.show()
```

(app.py)

```
from flask import Flask, render_template, request, jsonify
import cv2
import numpy as np
import tensorflow as tf
app = Flask(__name__)
# Load trained model
model = tf.keras.models.load_model("emotion_model.keras")
# Labels
labels = ['angry', 'fear', 'happy', 'neutral', 'surprise', 'sad', 'disgust']
# Emojis
emoji = {
    "angry": "😡",
    "fear": "😨",
    "happy": "😄",
    "neutral": "😐",
    "surprise": "😮",
    "sad": "😞",
    "disgust": "😬"
}
# Face detector
face = cv2.CascadeClassifier(
    cv2.data.haarcascades +
    "haarcascade_frontalface_default.xml"
)
@app.route("/")
def home():
    return render_template("index.html")
@app.route("/predict", methods=["POST"])
def predict():
    if "image" not in request.files:
        return jsonify({"success": False, "error": "Missing image file"}), 400
    file = request.files["image"]
    if file.filename == "":
        return jsonify({"success": False, "error": "Empty filename"}), 400
    img = np.frombuffer(file.read(), np.uint8)
    img = cv2.imdecode(img, cv2.IMREAD_COLOR)
    if img is None:
        return jsonify({"success": False, "error": "Invalid image"}), 400
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    faces = face.detectMultiScale(
        gray,
        scaleFactor=1.1,
        minNeighbors=6,
        minSize=(80,80)
```



```

)
if len(faces) == 0:
    return jsonify({"success": False})

x, y, w, h = max(faces, key=lambda f: f[2] * f[3])

roi = gray[y:y+h, x:x+w]
roi = cv2.resize(roi, (48,48))
roi = roi / 255.0
roi = roi.reshape(1,48,48,1)
pred = model.predict(roi, verbose=0)[0]
conf = float(np.max(pred))
label = labels[np.argmax(pred)]
return jsonify({
    "success": True,
    "emotion": label.title(),
    "emoji": emoji[label],
    "confidence": round(conf * 100, 2),
    "box": {
        "x": int(x),
        "y": int(y),
        "w": int(w),
        "h": int(h)
    }
})

if __name__ == "__main__":
    app.run(debug=True)

```

Appendix B: Screenshots

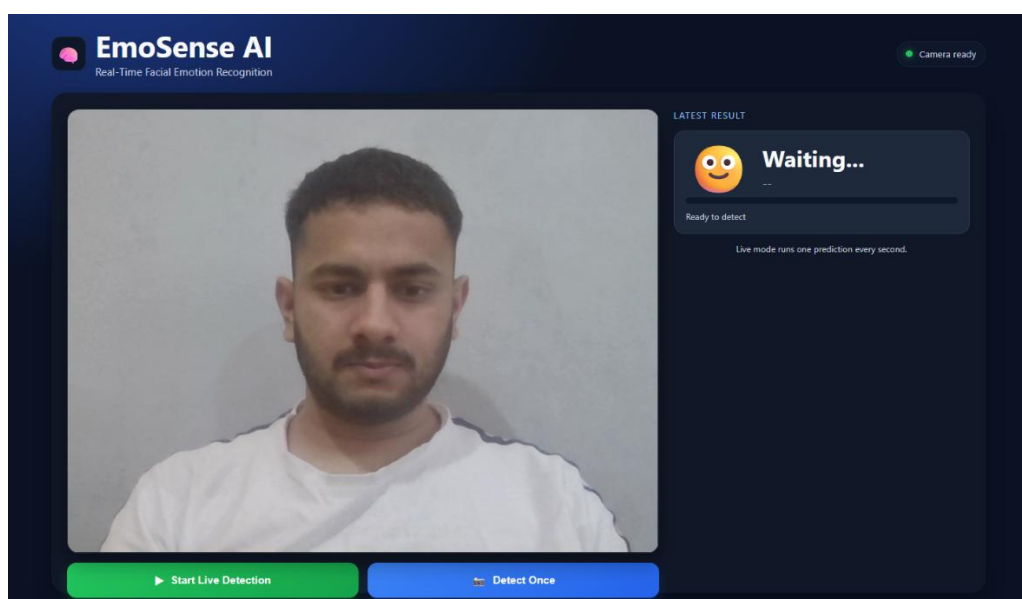


Figure B.1: User Interface of EmoSense AI

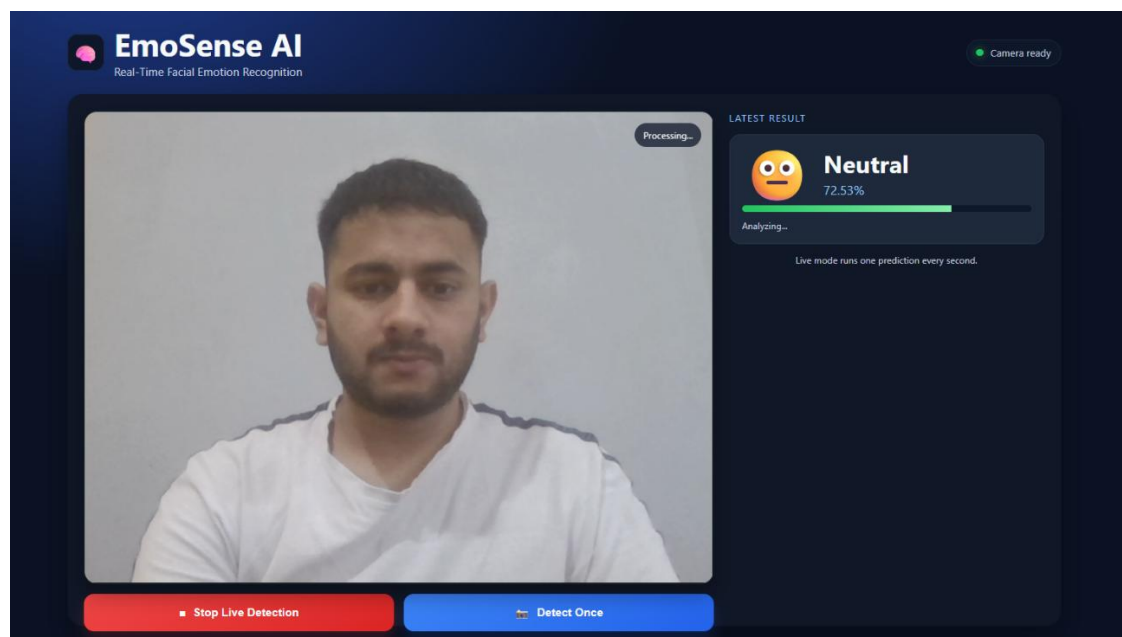


Figure B.2: Detection of Neutral Facial Expression

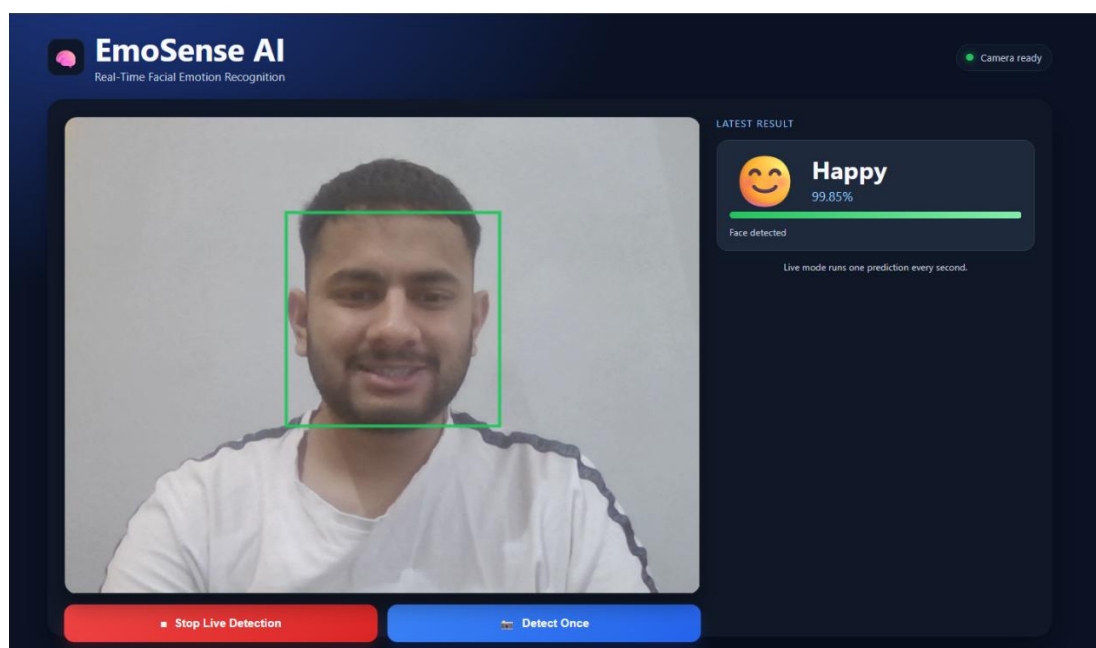


Figure B.3: Detection of Happy Emotion with High Confidence

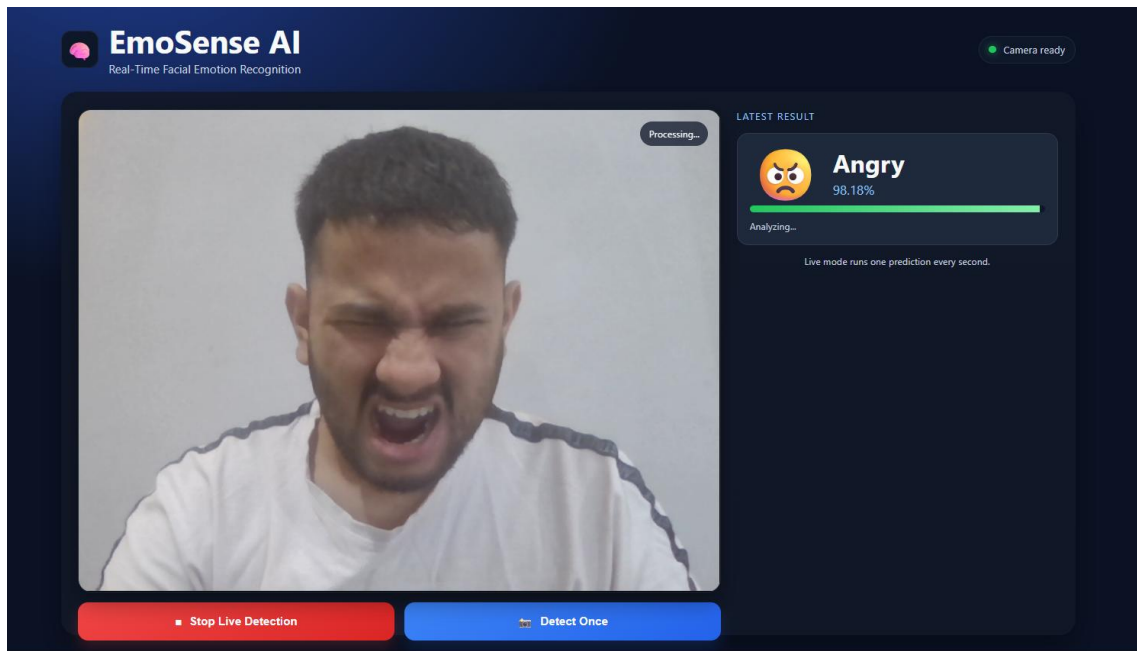


Figure B.4: Detection of Angry Emotion

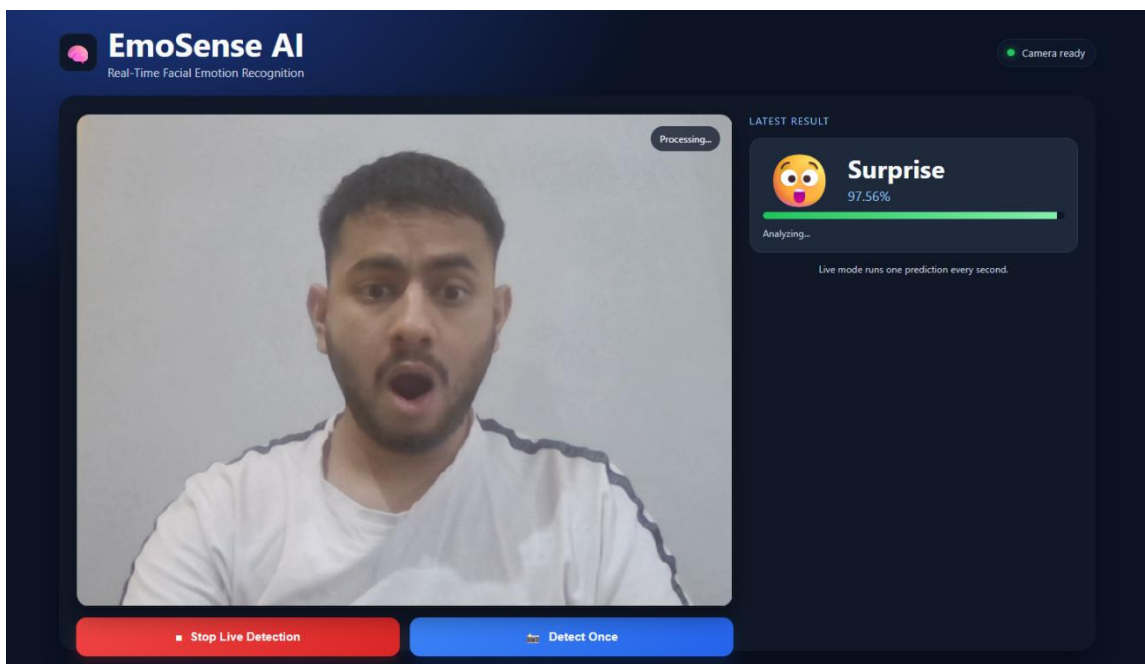


Figure B.5: Detection of Surprise Emotion

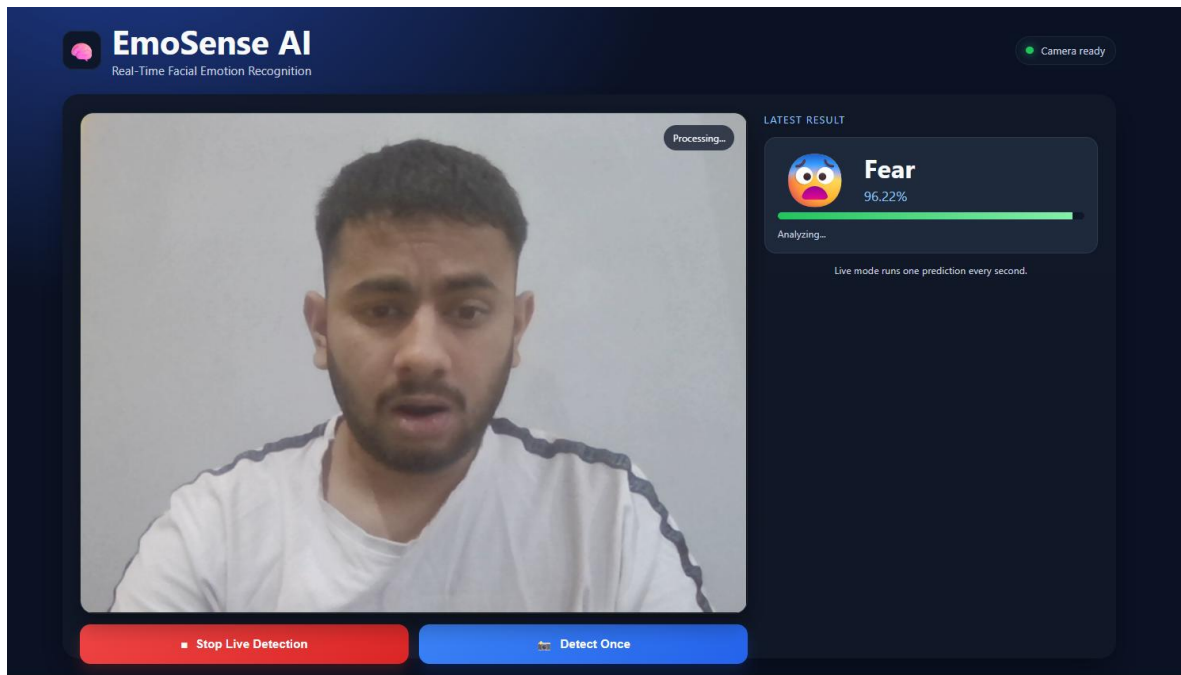


Figure B.6: Detection of Fear Emotion

Appendix C: System Requirements

C.1 Hardware Requirements

The minimum hardware configuration required to run the EmoSense AI system is as follows:

- Processor: Intel Core i5 or AMD Ryzen 5 (or higher)
- RAM: 8 GB or above
- Storage: Minimum 2 GB free disk space
- Camera: Integrated or external webcam
- GPU (Optional): NVIDIA GPU for faster training and inference

C.2 Software Requirements

The system is developed using the following software tools and libraries:

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.x
- Deep Learning Framework: TensorFlow and Keras

- Computer Vision Library: OpenCV
- Web Framework: Flask
- Frontend Technologies: HTML, CSS, JavaScript
- Browser: Google Chrome (recommended)
- Development Tools: VS Code or any Python IDE